



Securing understanding of equality

Task A: Maintaining equality in bar models

1) Complete each bar model so it maintains equality, given that $2a + 3 = b + 8$

a)

a	a	3	5
b	8		5

d)

a	a	3	b
b	b	8	

b)

a	a	3	a
b	8		a

e) e.g.

a	a	a	a	3	3
b	8		b	8	

c)

a	a	3	a	a	3	a	a	3
b	8		b	8		b	8	

or

a	a	3	a	a	3	a	a	3
b	8		a	a	3	a	a	3

2) The bar models are not drawn to scale. Given that $2c + 18 = 3x + 21$

Which bar models will maintain equality, and should be the same length?

Explain your reasoning.

a)

c	c	18	18
x	x	x	21

No, 18 is added to the top but $2x$ added to the bottom.

d)

c	c	c	c	18
x	x	x	x	x

No, $2c$ added to the top and $3x$ added to bottom. The whole expression hasn't been doubled.

b)

c	c	18	c	c
x	x	x	21	c

Yes, $2c$ added to both. These should be equal.

e)

c	c	c	c	18	18
x	x	x	x	x	21

Yes, both expressions have been doubled. These should be equal.

c)

c	c	c	c	18	18
x	x	x	21	c	c

Yes, $2c + 18$ added to both. These should be equal.

f)

c	c	x	18
x	x	x	21

Yes, x has been added to both. These should be equal.



Task B: Maintaining equality in equations

1) Fill in the blanks in these equations so they maintain equality.

a)

$$a = b$$

$$a + 8 = b \boxed{+ 8}$$

$$a - 2 = b \boxed{- 2}$$

b)

$$a = b$$

$$a \boxed{+ r + 5} = b + r + 5$$

$$a \boxed{+ r + 3} = b + r + 3$$

c)

$$a = b$$

$$\boxed{3a} = 3b$$

d)

$$3x + 4 = y + 1$$

$$3x + 4 - x = y + 1 \boxed{- x}$$

e)

$$3x + 4 = y + 1$$

$$\boxed{8(3x + 4)} = 8(y + 1)$$

f)

$$3x + 4 = y + 1$$

$$3x + 4 + (-5) = \boxed{y + 1 + (-5)}$$

$$\text{or } \dots = \boxed{y - 4}$$

g)

$$p + q = 10$$

$$\frac{p + q}{5} = \boxed{\frac{10}{5}}$$

$$\text{or } = \boxed{2}$$

h)

$$p + q = 10$$

$$\boxed{3(p + q)} = 30 \text{ or}$$

$$\boxed{3p + 3q} = 30 \text{ or}$$

$$\boxed{p + q + 20} = 30$$

i)

$$p + q = 10$$

$$p + q + p + q = \boxed{10 + p + q}$$

$$\text{or } = \boxed{20}$$

